

KOKPEKTY, Kazakhstan

WMO: 365350

Lat: 48.750N

Lon: 82.367E

Elev: 512

StdP: 95.32

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.7	-34.0	-39.2	0.1	-36.4	-36.6	0.1	-33.8	8.5	-16.5	7.7	-15.9	0.5	310	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	32.4	18.1	30.6	17.6	28.9	16.9	19.6	28.8	18.6	27.7	17.8	26.9	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.3	23.5	15.1	11.4	23.0	14.0	10.6	22.3	58.3	28.9	55.1	27.6	52.3	26.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.2	-39.4	35.3	3.3	1.7	-41.7	36.5	-43.6	37.5	-45.4	38.4	-47.8	39.6		
(5)			-39.4	21.8	3.3	1.5	-41.7	22.9	-43.6	23.7	-45.4	24.6	-47.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-20.5	-17.0	-7.8	5.7	13.1	19.0	20.7	19.1	12.3	4.2	-6.1	-16.1		
	DBStd	15.55	7.32	6.69	7.07	5.49	4.10	3.55	3.00	3.38	4.44	4.52	7.20	6.94		
	HDD10.0	3924	947	757	552	146	18	1	0	0	27	183	484	809		
	HDD18.3	6053	1205	990	810	381	166	34	11	31	186	437	734	1068		
	CDD10.0	1114	0	0	0	16	114	271	332	282	95	4	0	0		
	CDD18.3	203	0	0	0	0	4	54	85	55	4	0	0	0		
	CDH23.3	2712	0	0	0	4	133	701	933	805	136	0	0	0		
	CDH26.7	924	0	0	0	0	27	230	337	299	32	0	0	0		
(14) Wind	WSAvg	1.9	1.2	1.3	1.6	2.4	2.6	2.3	2.0	2.1	2.1	1.9	1.7	1.3		
(15) Precipitation	PrecAvg	315	27	20	15	17	32	30	35	20	17	23	38	36		
	PrecMax	544	79	78	67	47	147	90	133	75	62	52	103	88		
	PrecMin	176	4	1	1	0	1	0	0	0	0	0	0	6		
	PrecStd	77	17	15	13	12	27	20	29	18	15	15	25	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.3	-0.8	9.1	23.3	29.8	33.0	34.8	34.8	30.3	21.3	10.6	0.2		
		MCWB	-4.2	-2.3	5.3	13.8	17.3	18.1	19.1	19.0	17.0	12.1	5.5	-1.5		
	2%	DB	-5.2	-3.2	5.6	20.5	26.6	31.2	32.3	32.1	26.8	18.0	7.3	-2.4		
		MCWB	-6.0	-4.6	2.8	12.3	15.4	17.6	18.2	18.0	15.7	10.8	3.5	-3.6		
	5%	DB	-7.1	-5.0	3.5	18.0	24.5	29.4	30.3	30.1	24.6	15.0	4.6	-4.4		
		MCWB	-7.9	-6.2	1.1	10.9	14.6	16.9	17.7	17.4	14.9	9.1	1.9	-5.6		
	10%	DB	-9.1	-6.7	1.8	15.3	22.3	27.5	28.3	27.8	22.2	12.6	2.4	-6.0		
		MCWB	-9.9	-7.8	-0.3	9.2	13.7	16.3	17.0	16.7	13.8	7.6	0.2	-7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.9	-2.3	5.2	14.7	18.3	20.4	21.0	20.7	18.1	12.8	5.9	-1.4		
		MCDB	-3.4	-0.9	8.6	22.5	27.2	29.0	30.1	31.5	28.2	19.6	9.9	0.1		
	2%	WB	-5.9	-4.6	3.0	12.7	16.4	18.8	19.7	19.1	16.5	11.0	3.7	-3.5		
		MCDB	-5.3	-3.3	5.5	19.8	24.6	28.0	28.3	29.4	25.6	17.4	6.8	-2.5		
	5%	WB	-7.9	-6.2	1.2	11.1	15.1	17.9	18.8	18.1	15.0	9.3	2.0	-5.5		
		MCDB	-7.1	-5.1	3.3	17.3	23.2	27.0	27.2	28.2	24.0	14.5	4.6	-4.5		
	10%	WB	-9.8	-7.8	-0.2	9.5	14.0	17.0	18.0	17.2	13.8	7.8	0.3	-7.0		
		MCDB	-9.1	-6.7	1.8	14.9	21.5	25.8	26.4	26.6	21.9	12.2	2.4	-6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.7	11.5	12.2	14.1	14.0	13.2	14.9	15.8	12.7	9.3	10.4		
		MCDBR	9.3	11.2	10.2	17.7	17.7	16.8	16.2	18.7	19.3	17.8	11.6	8.7		
		MCWBR	8.8	10.3	8.0	11.0	9.3	7.2	6.5	8.2	10.7	11.9	8.7	7.9		
		MCDBR	9.2	10.8	9.7	16.3	16.0	15.4	14.1	16.6	18.5	16.6	11.2	8.5		
	5% WB	MCWBR	8.7	10.0	7.8	10.6	9.3	7.9	7.2	8.4	11.0	11.5	8.6	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.320	0.370	0.410	0.397	0.402	0.419	0.384	0.345	0.346	0.323	0.293		
	taud		2.080	2.023	2.026	2.120	2.219	2.277	2.242	2.318	2.374	2.280	2.161	2.063		
	Ebn at Noon		739	802	824	834	863	860	837	853	855	777	692	669		
	Edh at Noon		88	119	141	143	135	129	131	116	99	92	81	78		
(44) All-Sky Solar Radiation	RadAvg		1.69	2.83	4.15	5.10	6.45	6.82	6.35	5.85	4.49	2.60	1.46	1.25		
	RadStd		0.09	0.10	0.30	0.37	0.37	0.41	0.43	0.36	0.33	0.25	0.15	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-0.76	-0.83	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page